

- Tim Fuchs (formerly Tim Puhlfürß)
- PhD project in Informatics at Uni Hamburg, collaboration with EuXFEL and DASHH Graduate School
- Focus on knowledge exchange between software engineers and users (especially with **documentation**)
- Previous studies at EuXFEL:
 - Problem-focused: **Questionnaire** about challenges in knowledge exchange
 - Solution-focused: **AI-powered VS Code extension** to generate documentation comments in source code
 - Solution-focused: **AI-powered knowledge base** to accelerate information retrieval from issue trackers and documents
- Final study of PhD project:
 - Solution-focused: **AI-powered tool to support users during offline data analysis**

Research Questions

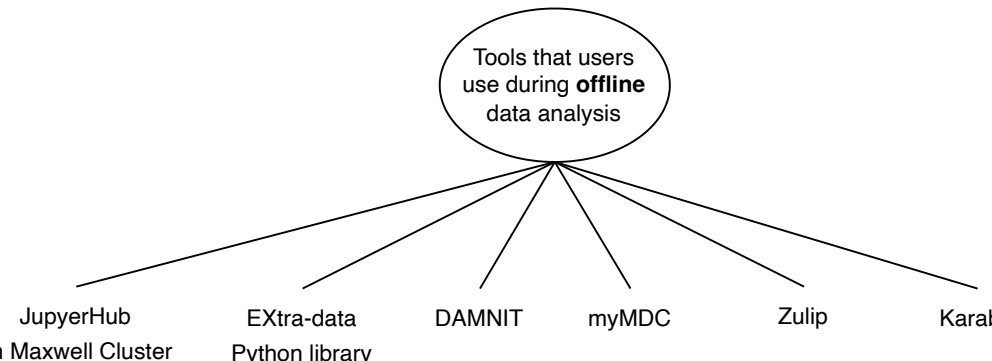
- RQ1: What **knowledge-related challenges** do EuXFEL users encounter during **offline** data analysis?
- RQ2: What are the **requirements of an AI agent** that addresses the challenges?
- RQ3: What **design recommendations** can be derived from deploying and evaluating the **AI agent in practice**?

Methodology

- **Design Science Research** with six (repeatable) activities:
 1. Problem identification and motivation (RQ1)
 2. Definition of the objectives for a solution (RQ2)
 3. Design and development (RQ3)
 4. Demonstration (RQ3)
 5. Evaluation (RQ3)
 6. Communication (paper publication)
- **Today: focus on problem identification (RQ1)**
 - RQ1: What **knowledge-related challenges** do EuXFEL users encounter during **offline data analysis**?
 - Method: focus group
 - Created artifacts: mind maps
- **Information on data protection:**
 - No personal information from individuals required
 - Demographic information that will be reported:
 1. "Focus group session was conducted with the Data Analysis group at EuXFEL."
 2. "[X] group members participated in the session."

Tools:

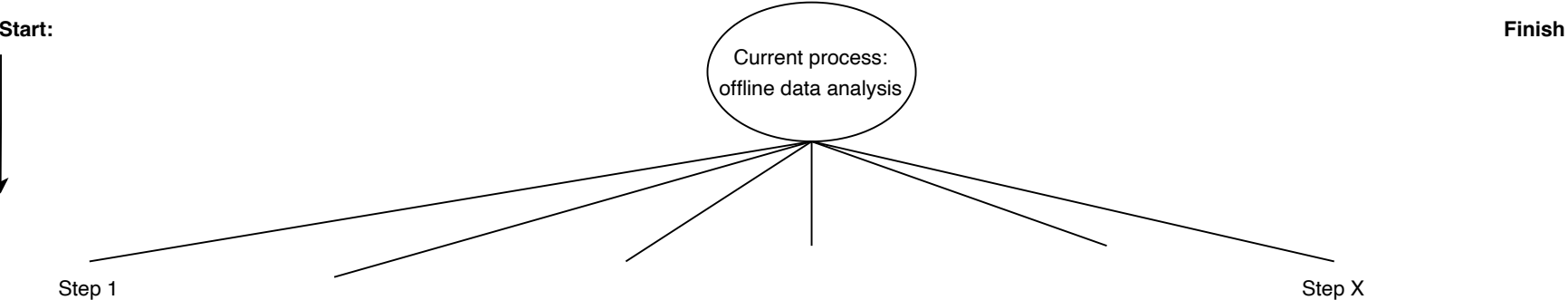
- What tools do users use during offline data analysis?



Current process:

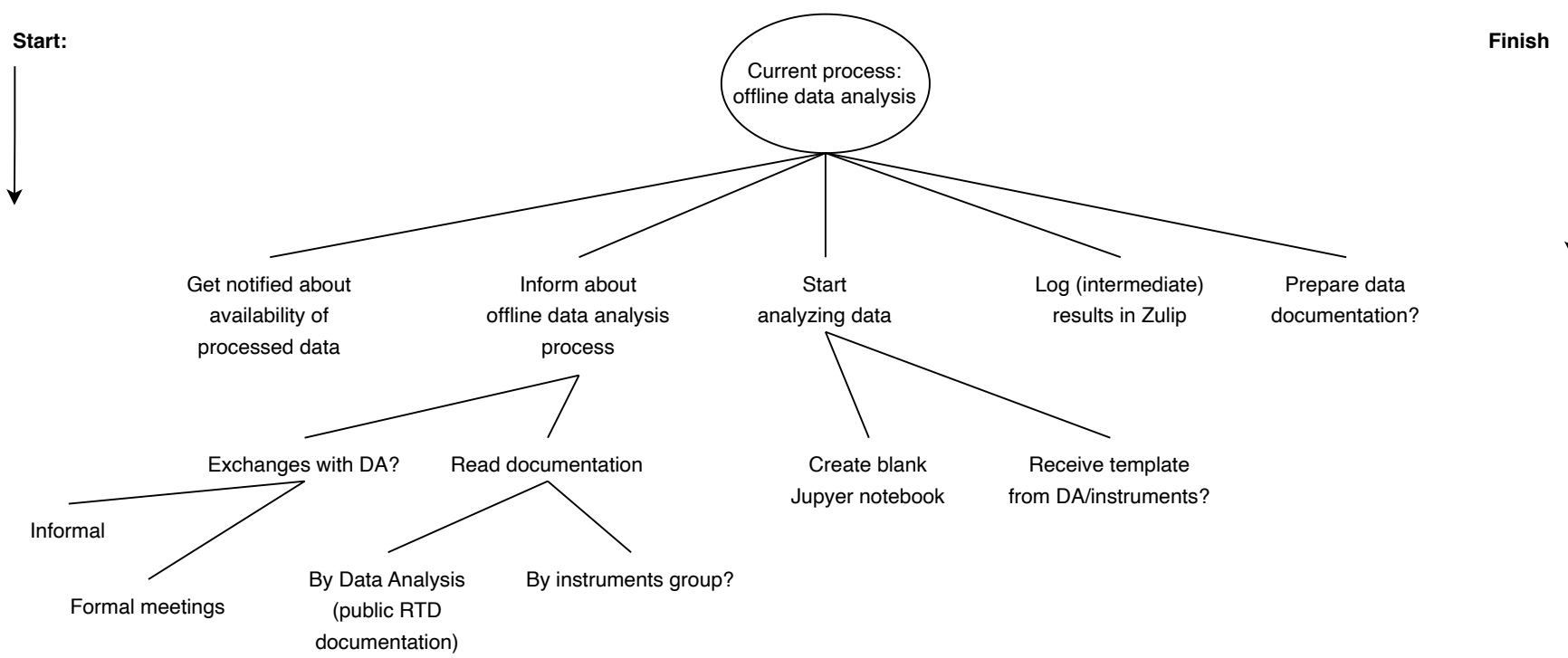
- What are the typical process steps of users to accomplish offline data analysis?
- How do users typically start their workflow when today tasks / requests from Endcustomers? Checklist needed?
- What artifacts do users typically produce during offline analysis? do they also prepare documentation for the data and their analysis artifacts (similar to a README file on GitHub.com etc)?
- How long do users typically analyze the data offline? I read between days and months.

Start:



Finish

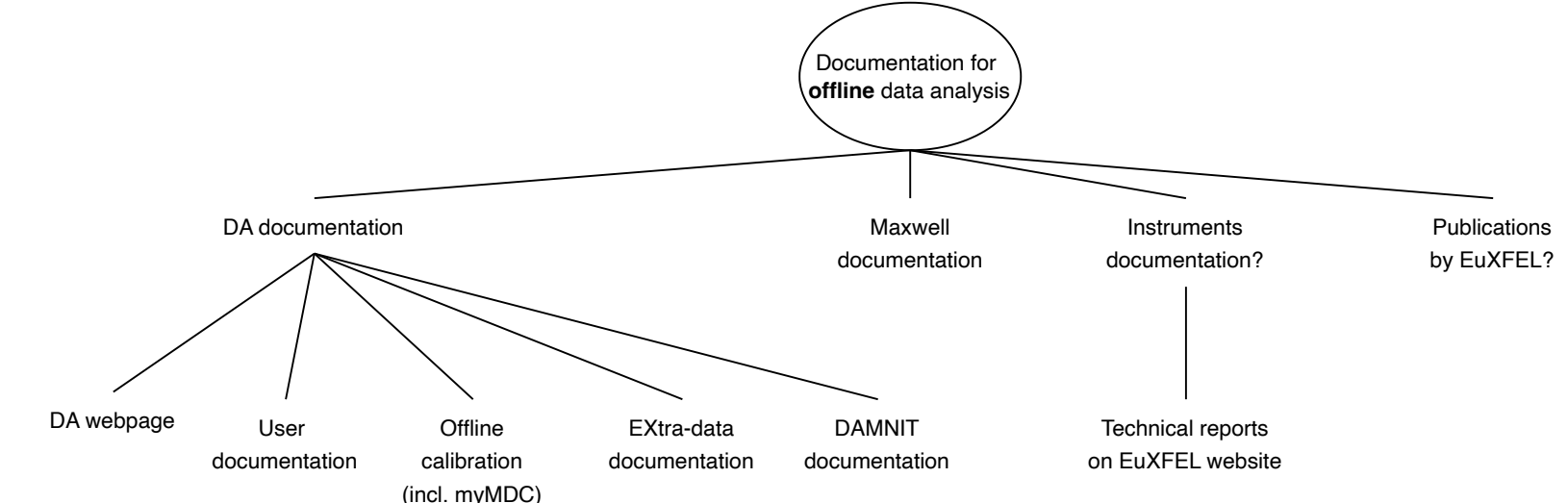
Start:



Finish

Documentation:

- What documentation do users require when analyzing the data (e.g., DA documentation, instrument documentation)?
- Where is this documentation located?
- From whom do users receive the documentation?



Locations:

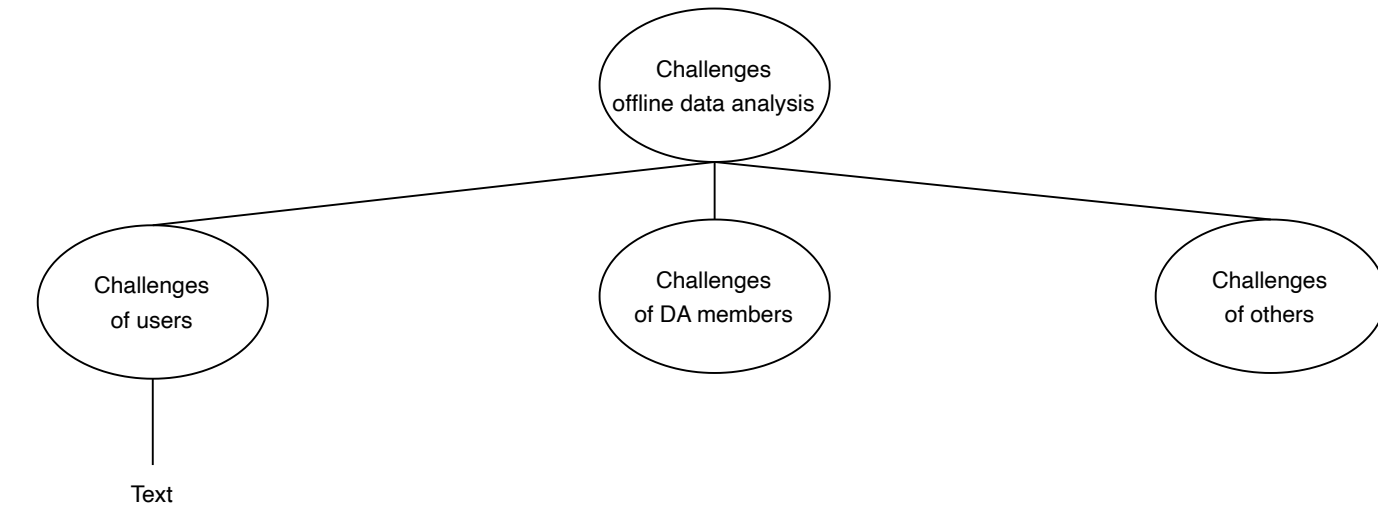
https://www.xfel.eu/organization/scientific_and_technical_groups/data_department/data_analysis/documentation_and_training_material/index_eng.html
<https://dataanalysis.pages.xfel.eu/user-documentation/>
<https://calibration.pages.xfel.eu/pycalibration/>
<https://extra-data.readthedocs.io/>
<https://damnit.readthedocs.io/>

<https://docs.desy.de/maxwell/documentation/>

https://www.xfel.eu/facility/instruments/spb_sfx/documentation/index_eng.html

Current challenges:

- What are knowledge-related challenges during offline data analysis (e.g., documentation, communication and support, clarity of tool features, requirements of users):
 - For users?
 - For DA members?
 - For other EuXFEL members, e.g., instruments groups?
- What are the biggest challenges at the moment?



Further information:

- Are there further resources regarding current challenges of users (e.g., previous surveys)?
- Is there anything else you would like to add?

